

Savar Verma

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EDUCATION

Indian Institute of Technology, Madras

Chennai, Tamil Nadu

Bachelor of Technology in Electrical Engineering

Batch of 2026

SKILLS

Languages: JavaScript, Python, C++, C, Bash, SQL

Frameworks & Tools: Node.js, React, Vite, FastAPI, Docker, Git, Linux

Cloud & Databases: AWS (EC2), DynamoDB, PostgreSQL, Redis

Architecture & Security: OAuth 2.1, PKCE, JWT, SSO, SRP, RBAC, REST APIs, Multithreading, System Design

WORK EXPERIENCE

Atomberg Intern

Feb 2026 – Present

Cloud & Backend Engineering (PPO Considerations)

Pune, India (Hybrid)

- Architected an **Identity Provider SaaS** on **EC2** using **Node.js** and **DynamoDB** to replace **Cognito**
- Implemented **OAuth 2.1** compliant flows with **authCode**, **PKCE**, and **JWTs** signed using **RS256** algorithm
- Engineered **multi-tenancy**, isolated **userpools**, and zero-knowledge **Secure Remote Password (SRP)** logins
- Integrated **Social SSO** (Google, Microsoft, and Apple) with dynamic configuration and provider toggling
- Developed a **React/Vite** admin and superadmin portal enforcing strict **Role-Based Access Model (RBAC)**
- Projected **97%** reduction in annual AWS expenditure, decreasing costs from **20 LPA** to **50K**

PROJECTS

High-Performance Scalable URL Shortener – *Python, React, PostgreSQL* – [[repo](#)] [[live](#)]

Nov 2025

- Achieved **sub-50ms** latency by architecting a **FastAPI** system with **Redis** cache-aside strategies
- Designed a collision-free **Base62** algorithm on **Serverless PostgreSQL** to handle millions of unique records
- Engineered **asynchronous** background tasks to track real-time geo-analytics without blocking user requests
- Built a responsive **React** and **Tailwind** dashboard to visualize live system telemetry and traffic metrics

Phonetic Memory Tool – *Python, SQLite, OpenAI, Prompt Engineering* – [[repo](#)]

Sep 2026

- Built a **phonetic memory layer** that learns a user's proper-name spellings from corrected **ASR** transcripts
- Engineered **Double Metaphone** retrieval with squashed n-grams and **in-SQL TF-IDF** ranking for **5ms** lookups
- Reserved the **LLM** as a final-check gate – temperature-0, strict **JSON** after deterministic local checks
- Benchmarked **500 cases**: **87.5%** intervention accuracy with **3.2%** false positives across **157 negatives**

Multi-Threaded CLI Image Editor – *C++* – [[repo](#)]

Jul 2025

- Designed and developed a command-line **image editor** for Linux in C++ for low-level **pixel manipulation**
- Leveraged C++ multithreading to achieve a **64% performance boost** over sequential execution
- Implemented strict in-memory processing to eliminate **I/O race conditions** during concurrent batch operations

OPEN SOURCE CONTRIBUTIONS

Suckless.org (dwm) Upstream Contributor – *C, X11* – [[patch 1](#)] [[patch 2](#)] [[commit](#)]

Jan 2026 – Present

- Contributed 2 upstream patches to **dwm** (**nocornermonocle**, **fullscreen**) to optimize **X11** window logic
- Integrated **EWMH compliance** via X11 Atoms to control external compositor rendering for specific layouts
- Engineered a **state-preserving** fullscreen toggle caching layout history to resolve view switching loops

RELEVANT COURSEWORK

Operating Systems, Probability and Detection Theory, Linear Algebra, Machine Learning Techniques